PowerGrid Predictive Maintenance

Capstone Project Report

Executive Summary

This project develops a leakage-aware predictive maintenance framework for PowerGrid Utilities Corporation. The supplied dataset contains 50,500 rows and 32 columns, with a 43.02% observed failure rate. Four models were compared. Random Forest was selected on overall ranking performance and high recall. The final holdout ROC-AUC was 0.908 and PR-AUC was 0.886 at a validation-selected threshold of 0.41.

1. Business Problem

PowerGrid faces equipment failures, outages, operational stress, aging assets, weather risk, and reactive maintenance. The analytical objective is to estimate failure probability and convert it into an interpretable maintenance priority.

2. Dataset Understanding

The workbook contains 50,500 observations and 32 columns. The target is grid_failure_flag. Exact duplicates: 500. Missingness is concentrated in oil_quality_score and inspection_score at roughly 10%. No reliable timestamp was found, so temporal validation was not possible.

3. Data Quality and Leakage

Exact duplicates were removed. Numeric missing values are median-imputed and categorical missing values are most-frequent imputed within the model pipeline. estimated_revenue_loss, regulatory_penalty_cost, and avg_outage_duration_minutes were excluded from prediction because they may be consequences of outages/failures. customers_served is used separately for business impact.

4. Exploratory Analysis

The analysis focuses on equipment condition, maintenance overdue days, load utilization, voltage fluctuation, storm risk, inspection and oil quality. These variables are treated as risk signals rather than causal claims. Generated charts are in outputs/figures.

5. Feature Engineering

The model uses defensible operational and condition variables available in the supplied data. The preprocessing pipeline performs imputation, scaling for numeric features, and one-hot encoding for categorical features. No synthetic rows or target values were created.

6. Modeling

Logistic Regression provides an interpretable baseline. Decision Tree provides rule-like explainability. Random Forest provides a robust ensemble. A linear SVM is included as the requested margin-based model. Class weighting is used instead of SMOTE because the observed failure rate is not an extreme minority class.

7. Model Evaluation

Evaluation emphasizes Precision, Recall, F1, ROC-AUC and PR-AUC. Recall is important because missed failures can lead to outages, while precision matters because excessive alarms consume maintenance capacity.

8. Model Comparison

Model Precision Recall F1 ROC-AUC PR-AUC Threshold Random Forest 0.740897 0.870428 0.800456 0.906451 0.881555 0.410 Decision Tree 0.746914 0.862678 0.800633 0.900320 0.870022 0.400 SVM 0.649175 0.816801 0.723404 0.830670 0.790319 0.470 Logistic Regression 0.647475 0.822691 0.724642 0.830680 0.790316 0.415

9. Selected Model

Random Forest was selected. It provides the strongest overall validation ranking and a strong balance of precision, recall, F1, ROC-AUC and PR-AUC. The operational threshold selected on validation data was 0.41, rather than assuming 0.50.

10. Risk Scoring

Failure probability is mapped to Low (<30%), Medium (30–60%), High (60–80%), and Critical (>=80%). Maintenance priority is a separate business concept combining predictive probability with normalized customers served.

11. High-Risk Asset Prioritization

The generated file outputs/high_risk_assets.csv contains the actual scored assets. The top-ranked assets should be reviewed by maintenance and engineering teams; model output is a decision-support tool, not a substitute for engineering judgment.

12. Recommendations

Inspect Critical-risk assets first; prioritize poor-condition and overdue-maintenance assets; monitor high-load and unstable-voltage conditions; incorporate weather exposure into operational planning; integrate scores into maintenance workflows; track model outcomes and false alarms.

13. Limitations

No timestamp means no genuine future-period validation. The dataset does not establish causality. Real-world deployment requires validation against live utility data, data-quality monitoring, calibration, and model-drift monitoring.

14. Future Scope

Add timestamps, IoT telemetry, time-series forecasting, survival analysis, anomaly detection, automated maintenance alerts, MLOps, and integration with asset-management systems when data supports them.

Conclusion

The project demonstrates a reproducible, interpretable predictive-maintenance workflow that separates failure risk from business impact and supports proactive maintenance prioritization.